

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-1

WINTER SEMESTER, 2024-2025

DURATION: 35 MINUTES

FULL MARKS: 15

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Suppose, R and k are arbitrary numbers such that $R > 0$ and $k \neq 0$. Prove that the Newton-Raphson formula for finding the k th root of R , which is mathematically denoted as $\sqrt[k]{R}$, is as shown in Equation-1. 3
(CO2)
(PO1)

$$x_{i+1} = \frac{1}{k} \left[(k-1)x_i + \frac{R}{x_i^{k-1}} \right] \quad (1)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \sqrt[k]{R} \\ \Rightarrow x^k &= R \\ \Rightarrow x^k - R &= 0 \end{aligned}$$

We take $f(x) = x^k - R$ and so $f'(x) = kx^{k-1}$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^k - R}{kx_i^{k-1}} \\ &= \frac{kx_i^k - x_i^k + R}{kx_i^{k-1}} \\ &= \frac{(k-1)x_i^k + R}{kx_i^{k-1}} \\ &= \frac{1}{k} \left[\frac{(k-1)x_i^k + R}{x_i^{k-1}} \right] \\ &= \frac{1}{k} \left[(k-1)x_i + \frac{R}{x_i^{k-1}} \right] \quad \square \text{ Q.E.D.} \end{aligned}$$

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 2 points for correct derivation.

2. a) What are significant digits? What is the relationship between the number of significant digits that are at least correct (m) in an approximation and the relative approximate error (ϵ_a)? 1
(CO2)
(PO1)

Solution:

Significant figures (also known as the significant digits, precision, or resolution) of a number in positional notation are digits in the number that are reliable and necessary to indicate the quantity of something. These are the digits that are known reliably in a reported number. The relationship between m and ϵ_a is,

$$|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$$

Rubric:

- 0.5 point for a correct definition. (Doesn't need to be exactly as written above.)
- 0.5 point for the correct relationship.

- b) Determine the *minimum* number of significant digits in each of these numerical values — 3
(CO4)
(PO1)
- | | | |
|--------------|------------|-------------------------------|
| (a) 69.0 | (b) 69,420 | (c) 6.94200×10^{911} |
| (d) 69.00420 | (e) 069420 | (f) 0.069420 |

Solution:

We need to determine the *minimum* number of significant digits.

- (i) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.
- (ii) 69,420 has 4 significant digits, since trailing 0's in an integer may or may not be significant.
- (iii) 6.94200×10^{911} has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.
- (iv) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.
- (v) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.
- (vi) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.

Rubric:

- 0.5 point for each correct answer. (The reasons are not required.)

3. Draw a rough sketch of the graph of two functions $f_1(x)$ and $f_2(x)$ given that — 3
(CO3)
(PO2)
- For $f_1(x)$, False Position method performs better than the Bisection method.
 - For $f_2(x)$, Bisection method performs better than the False Position method.

Here, "*performs better*" means that it takes a lesser number of iterations to converge to the root. You have the liberty to choose the initial guesses $\langle x_l, x_u \rangle$ as well.

Solution:

$f_1(x)$ may look something like,

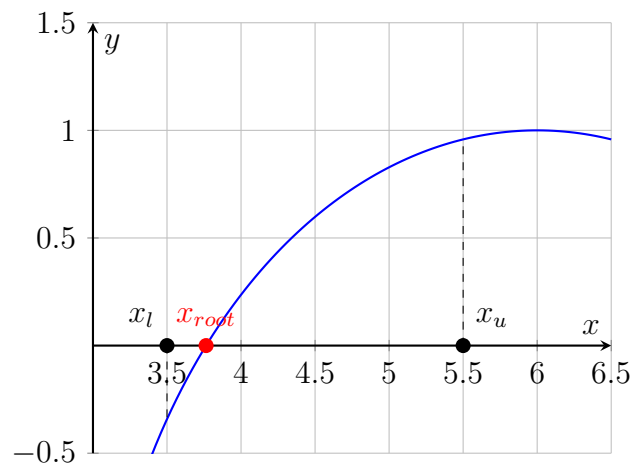


Figure 1: Graph of $f_1(x)$.

$f_2(x)$ may look something like,

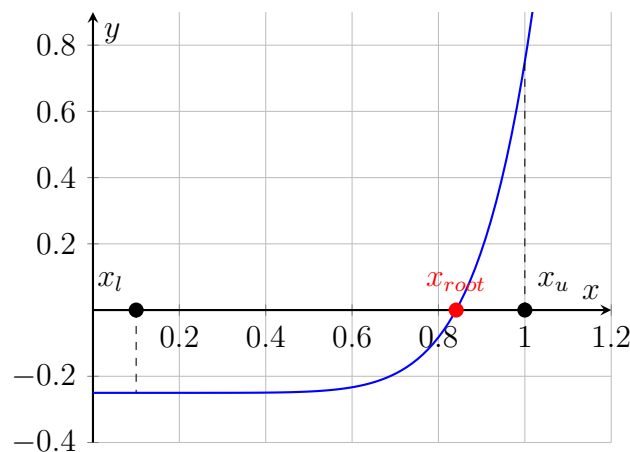


Figure 2: Graph of $f_2(x)$.

Rubric:

- 1.5 points for a correct $f_1(x)$ graph sketch.
- 1.5 points for a correct $f_2(x)$ graph sketch.

4. The *Plastic Ratio*, often denoted by the Greek letter ρ (rho), is the ratio involving three quantities a , b , and c , such that $a > b > c > 0$, if they satisfy the equality $\frac{a}{b} = \frac{b+c}{a} = \frac{b}{c}$. The constant ρ is equal to the unique real root of the cubic equation $x^3 = x + 1$. The numerical value of that root is,

$$x_{root} = \rho = \sqrt[3]{\frac{9 + \sqrt{69}}{18}} + \sqrt[3]{\frac{9 - \sqrt{69}}{18}} = 1.324717957 \dots$$

5
(CO2)
(PO1)

Apply the *Regula Falsi* method, also known as the False Position method, to estimate the value of the plastic ratio ρ . The initial upper and lower guesses are $x_l = -1$ and $x_u = 2$ respectively. Conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m), at the end of each iteration. Fill out the following table after performing the necessary calculations.

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	-1	2		—	—	
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	-1	2	-0.5	—	—	-0.625
2	-0.5	2	-0.2222	125	0	-0.7887
3	-0.2222	2	0.0805	376.0248	0	-1.08
4	0.0805	2	0.4215	80.9015	0	-1.3466

Rubric:

- 0.25 point for each correct answer.

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QUIZ-2

DURATION: 40 MINUTES

WINTER SEMESTER, 2024-2025

FULL MARKS: 15

Math 4543: Numerical Methods

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Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Euler's identity, which is $e^{i\pi} + 1 = 0$, is sometimes referred to as the most beautiful equation in mathematics. It is a special case of Euler's formula 6
- $$e^{ix} = \cos(x) + i \sin(x) \quad \text{(CO2)}$$
- (PO1)

when evaluated for $x = \pi$. One of the many proofs of this formula involves utilizing the Taylor expansions of its constituent transcendental functions.

The Taylor series for a function $f(x)$ is –

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of –

- (i) e^x
- (ii) $\cos(x)$
- (iii) $\sin(x)$

Then use these three series to prove Euler's formula.

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(e^x) = e^x$, $\frac{d}{dx}(\sin(x)) = \cos(x)$, and $\frac{d}{dx}(\cos(x)) = -\sin(x)$.

- (i) Now, for $f(x) = e^x$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = e^x, f(0) = e^0 = 1; f'(x) = e^x, f'(0) = e^0 = 1; f''(x) = e^x, f''(0) = e^0 = 1; f'''(x) = e^x, f'''(0) = e^0 = 1; f^{(4)}(x) = e^x, f^{(4)}(0) = e^0 = 1;$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f''''(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \frac{h^4}{4!} + \dots \\
 \Rightarrow f(x) &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \\
 \Rightarrow e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots
 \end{aligned}$$

(ii) Now, for $f(x) = \cos(x)$, let's calculate the derivatives at the point $x = 0$.

$$\begin{aligned}
 f(x) &= \cos(x), f(0) = \cos(0) = 1; f'(x) = -\sin(x), f'(0) = -\sin(0) = 0; \\
 f''(x) &= -\cos(x), f''(0) = -\cos(0) = -1; f'''(x) = \sin(x), f'''(0) = \sin(0) = 0; \\
 f''''(x) &= \cos(x), f''''(0) = \cos(0) = 1;
 \end{aligned}$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f''''(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 1 + 0 - \frac{h^2}{2!} + 0 + \frac{h^4}{4!} - \dots \\
 \Rightarrow f(h) &= 1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \dots \\
 \Rightarrow f(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \\
 \Rightarrow \cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots
 \end{aligned}$$

(iii) Now, for $f(x) = \sin(x)$, let's calculate the derivatives at the point $x = 0$.

$$\begin{aligned}
 f(x) &= \sin(x), f(0) = \sin(0) = 0; f'(x) = \cos(x), f'(0) = \cos(0) = 1; f''(x) = -\sin(x), \\
 f''(0) &= -\sin(0) = 0; f'''(x) = -\cos(x), f'''(0) = -\cos(0) = -1; f''''(x) = \sin(x), \\
 f''''(0) &= \sin(0) = 0; \text{ Using the first 5 terms of the Taylor series,}
 \end{aligned}$$

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f''''(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$f(0 + h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow f(h) = 0 + h + 0 - \frac{h^3}{3!} + 0 + \dots$$

$$\Rightarrow f(h) = h - \frac{h^3}{3!} + \dots$$

$$\Rightarrow f(x) = x - \frac{x^3}{3!} + \dots$$

$$\Rightarrow \sin(x) = x - \frac{x^3}{3!} + \dots$$

Now,

$$\text{L.H.S} = e^{ix}$$

$$= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \dots \quad [\text{Since, } i^2 = -1, i^3 = -i, \text{ and } i^4 = 1.]$$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i \left(x - \frac{x^3}{3!} + \dots\right) \quad [\text{Separating the real and complex terms.}]$$

$$= \cos(x) + i \sin(x)$$

$$= \text{R.H.S.} \quad \square \text{ Q.E.D.}$$

Rubric:

- 1.5 points for the correct series of e^x .
- 1.5 points for the correct series of $\cos(x)$.
- 1.5 points for the correct series of $\sin(x)$.
- 1.5 points for the correct proof of Euler's formula.

2. Differentiate between *Interpolation* and *Extrapolation*. With suitable real-world examples, briefly write when each of the methods is applicable.

2
(CO3)
(PO1)

Solution:

Any valid conceptual difference will suffice. There should be at least 1 valid real-world example that is used to explain the applicability. Some differences are –

Criteria	Interpolation	Extrapolation
Position of newly estimated points	Within the distribution of the known data points.	Outside the distribution of the known data points.
Likelihood of proper estimation	Has a greater likelihood of obtaining a valid estimate, since the pattern of the data points on either side of the newly estimated point is known.	Has a lesser likelihood of obtaining a valid estimate, since it assumes that the observed trend continues for values of x outside the range of the given data points.
Applicability	• Computer Graphics – For constructing B-splines, smooth surfaces, etc. • Carpentry – For drawing curves on wooden surfaces using different apparatus e.g. the Spline tool. • Digital Image Processing – For increasing the resolution of images. • Electronics – For plotting transfer functions or calibration curves of electronic components e.g. Thermistors.	

Rubric:

- 1 point for a valid conceptual difference.
- 0.5 point for each valid real-world application.

3. A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Table-1 portrays multiple recorded observations involving a thermistor.

4
(CO2)
(PO1)

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the Lagrangian method of interpolation and a second order polynomial.

Table 1: Temperature T of a thermistor as a function of its resistance R for Question-3.

Resistance, R (Ω)	Temperature, T ($^{\circ}C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Solution:

The 2nd order Lagrangian interpolant is –

$$T(R) = \sum_{i=0}^2 L_i(R)T(R_i)$$

$$= L_0(R)T(R_0) + L_1(R)T(R_1) + L_2(R)T(R_2)$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.

Now, let's calculate the Lagrangian weights.

$$L_0(R) = \prod_{\substack{j=0 \\ j \neq 0}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 636}{911.3 - 636} \right) \times \left(\frac{754.8 - 451.1}{911.3 - 451.1} \right) = 0.284$$

$$L_1(R) = \prod_{\substack{j=0 \\ j \neq 1}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{636 - 911.3} \right) \times \left(\frac{754.8 - 451.1}{636 - 451.1} \right) = 0.933$$

$$L_2(R) = \prod_{\substack{j=0 \\ j \neq 2}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{451.1 - 911.3} \right) \times \left(\frac{754.8 - 636}{451.1 - 636} \right) = -0.218$$

Now that we have the weights, we can obtain the value of $T(754.8)$.

$$T(754.8) = 0.284 \times T(911.3) + 0.933 \times T(636) + (-0.218) \times T(451.1)$$

$$= 0.284 \times 30.131 + 0.933 \times 40.12 + (-0.218) \times 50.128$$

$$\approx 35.088^{\circ}C$$

Rubric:

- 1 point for the correct polynomial representation.
- 0.5 point for choosing the correct nearest points.
- 0.5 point for each Lagrangian weight $L_i(R)$ value.
- 1 point for the correct answer.

4. Prove that a polynomial of degree n or less that passes through $n + 1$ data points is unique.

3
(CO4)
(PO1)

Solution:

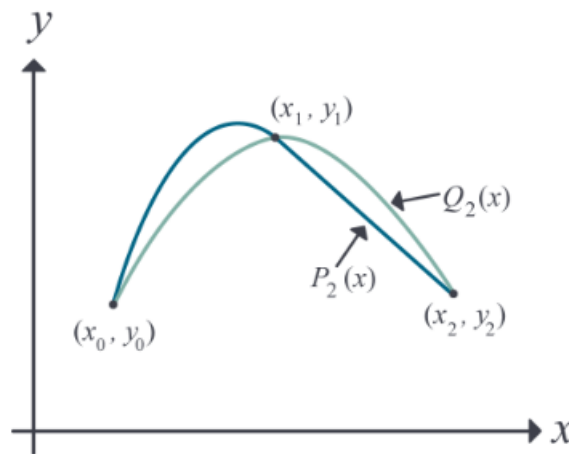


Figure 1: Interpolating polynomials through three points.

Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n + 1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n + 1$ data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$. Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n + 1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \dots, n\}$$

Hence

$$\begin{aligned} R_n(x_i) &= P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \dots, n\} \\ \Rightarrow R_n(x_i) &= 0 \end{aligned}$$

The n th order polynomial $R_n(x)$ has $n + 1$ zeros. A polynomial of order n can have $n + 1$ zeros only if it is identical to a zero polynomial, that is,

$$\begin{aligned} R_n(x) &\equiv 0 \\ \Rightarrow P_n(x) &\equiv Q_n(x) \end{aligned}$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. $\square$ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 1 point for the final conclusion.

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QUIZ-3

DURATION: 35 MINUTES

WINTER SEMESTER, 2024-2025

FULL MARKS: 15

Math 4543: Numerical Methods

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Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. The zero y -intercept version of the linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is 3 + 2
- $$y = a_1 x$$
- (CO4)
-
- (PO1)

where a_1 is the only parameter of the regression model.

- i. Derive the formula for finding the optimal value of a_1 .
- ii. Prove that the value of a_1 obtained using the formula derived in Question-1(i) is unique and corresponds to the absolute minimum of the optimization criterion used.

Solution:

- i. The residual error for each given data point is,

$$E_i = y_i - a_1 x_i$$

So the sum of the square of the residual errors is,

$$S_r = \sum_{i=1}^n E_i^2 = \sum_{i=1}^n (y_i - a_1 x_i)^2$$

To find a_1 , we look for the value of a_1 for which S_r is the absolute minimum. At first, let's do the first derivative test.

$$\begin{aligned} \frac{dS_r}{da_1} &= 2 \sum_{i=1}^n (y_i - a_1 x_i)(-x_i) = 0 \\ \Rightarrow -2 \sum_{i=1}^n y_i x_i + 2 \sum_{i=1}^n a_1 x_i^2 &= 0 \Rightarrow -2 \sum_{i=1}^n y_i x_i + 2a_1 \sum_{i=1}^n x_i^2 = 0 \\ \Rightarrow a_1 &= \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2} \end{aligned}$$

Rubric:

- 0.5 point for the correct equation of E_i .
- 1 point for the correct equation of S_r .
- 1.5 points for the correct derivation of a_1 .

- ii. To prove the uniqueness of a_1 , we need to prove that its value corresponds to the absolute minimum of S_r . For this, let's invoke the second derivative test.

$$\begin{aligned}\frac{d^2 S_r}{da_1^2} &= \frac{d}{da_1} \left(\frac{dS_r}{da_1} \right) \\ &= \frac{d}{da_1} \left(2 \sum_{i=1}^n (y_i - a_1 x_i)(-x_i) \right) \\ &= \frac{d}{da_1} \left(\sum_{i=1}^n (-2x_i y_i + 2a_1 x_i^2) \right) \\ &= \sum_{i=1}^n 2x_i^2 > 0\end{aligned}$$

by pragmatically assuming that all the x_i values are not zero at the same time. Hence, S_r yields the absolute minimum value at that value of a_1 . So the value of a_1 that is obtained using the formula in Question-1(i) is unique. $\square$ Q.E.D.

Rubric:

- 1.5 points for the correct second derivative test.
- 0.5 point for the correct conclusion.

2. Suppose you are analyzing the catalytic efficiency of an enzyme to determine the maximum reaction velocity $v_{\max}$ and the Michaelis constant K_m . The kinetics of the reaction can be represented by the Saturation Growth model (Michaelis-Menten equation):

4
(CO2)
(PO1)

$$v = \frac{v_{\max} S}{K_m + S}$$

where v is the initial reaction velocity and S is the substrate concentration. The values of reaction velocity v and the respective substrate concentration S that you have measured experimentally are shown in the table below –

Substrate Conc., S (mM)	Reaction Velocity, v ($\mu\text{mol} \cdot \text{min}^{-1}$)
1	1.7
2	2.9
5	5.0
10	6.7
20	8.0
40	8.9

Table 1: Experimental kinetic data for the enzymatic reaction.

Determine the value of the maximum velocity $v_{\max}$ and the Michaelis constant K_m using the data provided in Table-1. You are allowed to apply data transformation (linearization).

Solution:

Let's linearize the data at first. Taking the reciprocal of both sides of the given Saturation Growth model (Michaelis-Menten equation), we get the Lineweaver-Burk equation:

$$\frac{1}{v} = \frac{K_m}{v_{\max}} \frac{1}{S} + \frac{1}{v_{\max}}$$

Let, $z = \frac{1}{v}$, $w = \frac{1}{S}$, $a_1 = \frac{K_m}{v_{\max}}$, and $a_0 = \frac{1}{v_{\max}}$; implying that $v_{\max} = \frac{1}{a_0}$ and $K_m = a_1 \cdot v_{\max}$.

The straight line that we end up with is

$$z = a_0 + a_1 w$$

Using the calculator, we obtain the following values for the $\hat{n} = 6$ transformed data points ($1/S$ and $1/v$).

$$\sum_{i=1}^{\hat{n}} w_i = 1.875, \sum_{i=1}^{\hat{n}} z_i = 1.5197, \sum_{i=1}^{\hat{n}} w_i z_i = 0.8246, \sum_{i=1}^{\hat{n}} w_i^2 = 1.3031$$

Now, let's apply the formulae for determining the corresponding linear regression parameters a_0 and a_1 .

$$a_1 = \frac{\hat{n} \sum_{i=1}^{\hat{n}} w_i z_i - \sum_{i=1}^{\hat{n}} w_i \sum_{i=1}^{\hat{n}} z_i}{\hat{n} \sum_{i=1}^{\hat{n}} w_i^2 - \left(\sum_{i=1}^{\hat{n}} w_i \right)^2} = 0.4876, a_0 = \frac{\sum_{i=1}^{\hat{n}} z_i}{\hat{n}} - a_1 \frac{\sum_{i=1}^{\hat{n}} w_i}{\hat{n}} = 0.1009$$

Hence, we obtain

$$\begin{aligned} v_{\max} &= \frac{1}{0.1009} \\ &= 9.9108 \end{aligned}$$

$$K_m = 0.4876 \times 9.9108 = 4.8325$$

Plugging these values into the given Saturation Growth model equation, we get,

$$v = \frac{9.9108S}{4.8325 + S}$$

Rubric:

- 1 point for the correct linear representation.
- 1 point for the correct a_1 formula.
- 0.5 point for the correct a_1 value.
- 1 point for the correct a_0 formula.
- 0.5 point for the correct a_0 value.
- 0.5 point for the correct $v_{\max}$ value.
- 0.5 point for the correct K_m value.

3. Observe the lightly shaded solid object portrayed in Figure 1.

Its base is the region enclosed by a curve $y = \sqrt{x + x \ln x}$, the x -axis, the straight line $x = 1$, and the straight line $x = 2$. When this solid object is cut by a plane perpendicular to the x -axis and the xy -plane, the resulting cross-section is an equilateral triangle (as indicated by the darkly shaded cross-sectional area).

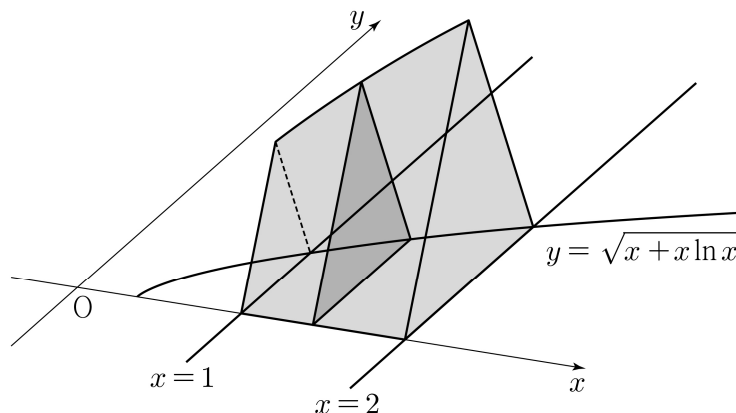


Figure 1: A solid geometric object.

- a) Formulate an integrand function $f(x)$ whose integral value within the limits $[1, 2]$ will yield the volume of the solid object.

1.5
(CO1)
(PO1)

Solution:

There are 2 key observations here—

- The cross-sections of the object are equilateral triangles.
- The sides of the triangles are gradually increasing according to the increasing nature of the $y = \sqrt{x + x \ln x}$ curve over the interval $[1, 2]$.

We know, that the area of an equilateral triangle having a side of length s is $\frac{\sqrt{3}}{4}s^2$. Here, for a particular value of x , the side length of an infinitesimally thin triangular cross-sectional slice is $\sqrt{x + x \ln x}$. So, the area of that triangular slice will be $\frac{\sqrt{3}}{4} \left(\sqrt{x + x \ln x} \right)^2$. All of these infinitesimally thin triangular slices within the limits $[1, 2]$ together constitute the entire volume of the solid object, *ergo* the integrand is,

$$f(x) = \frac{\sqrt{3}}{4}(x + x \ln x)$$

Rubric:

- 1.5 points for a correct integrand function.

- b) Using the Composite Trapezoidal Rule, with $n = 8$ segments, approximate the volume of the solid object.

4.5
(CO2)
(PO1)

Solution:

We can approximate the integral value of a function $f(x)$ using composite trapezoidal rule with n segments, each of width $h = \frac{b-a}{n}$, between the lower-limit a and the upper-limit b as follows,

$$\int_a^b f(x)dx = \frac{h}{2} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a + ih) \right\} + f(b) \right]$$

Here, $n = 8$ and $h = \frac{2-1}{8} = 0.125$. Let $(x_i = a + ih)$ for $(i = 0, \dots, 8)$. The x_i values and their corresponding functional values $f(x_i) = \frac{\sqrt{3}}{4}(x_i + x_i \ln x_i)$ are:

i	x_i	$f(x_i)$
0	1.000	0.433013
1	1.125	0.544519
2	1.250	0.662033
3	1.375	0.784994
4	1.500	0.912885
5	1.625	1.045330
6	1.750	1.182000
7	1.875	1.322615
8	2.000	1.466311

$$\begin{aligned} \int_1^2 f(x) dx &\approx \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^7 f(x_i) + f(x_8) \right] \\ &= \frac{0.125}{2} (0.433013 + 2 \times 6.454376 + 1.466311) \\ &\approx 0.925505 \end{aligned}$$

Rubric:

- 1 point for the correct composite trapezoidal rule formula.
- 2.5 points for the correct application of the formula.
- 1 point for the correct (or an approximately correct) answer.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-4

DURATION: 35 MINUTES

WINTER SEMESTER, 2024-2025

FULL MARKS: 15

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. State the fundamental theorems of calculus. Applying these theorems, explain how Euler's method is used to approximate the value of a definite integral. 3 + 2
(CO4)
(PO1)

Solution:

The fundamental theorems of calculus are –

- i. Let f be a continuous real-valued function defined on a closed interval $[a, b]$. Let F be the function defined, for all x in $[a, b]$ by

$$F(x) = \int_a^x f(t)dt$$

Then F is uniformly continuous on $[a, b]$ and differentiable on the open interval (a, b) , and $F'(x) = f(x)$ for all x in (a, b) so F is an antiderivative of f .

- ii. If f is Riemann integrable on $[a, b]$ then

$$\int_a^b f(x)dx = F(b) - F(a)$$

The idea is to rewrite the integral as the solution of an ODE,

$$\frac{dy}{dx} = f(x), y(a) = 0, \text{ using the 1st Fundamental theorem}$$

where then $y(b)$ gives us the value of the integral $\int_a^b f(x)dx$, according to the 2nd Fundamental theorem. We can use Euler's method to approximate the value of the dependent variable y at the discrete point $x = b$.

Rubric:

- 1.5 points each correct fundamental theorem.
- 2 points for the correct explanation.

2. Compare and contrast the following numerical algorithms for solving 1st order ODEs – Euler’s method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method.

2
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Euler	Runge-Kutta 2nd Order	Runge-Kutta 4th Order
Formula	$y_{i+1} = y_i + f(x_i, y_i)h$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2)h, k_1 = f(x_i, y_i), k_2 = f(x_i + ph, y_i + qk_1h)$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h$
Terms of the Taylor series	First 2 terms.	First 3 terms.	First 5 terms.
Order of the Taylor polynomial	1st order – which is why it’s also called the Runge-Kutta 1st order method.	2nd order – which is why it’s called the Runge-Kutta 2nd order method.	4th order – which is why it’s also called the Runge-Kutta 4th order method.
Local Truncation Error	$O(h^2)$	$O(h^3)$	$O(h^5)$
Global Truncation Error	$O(h)$	$O(h^2)$	$O(h^4)$
Variant Approaches	None.	Has 3 different versions – Heun’s method, Midpoint method, and Ralston’s method	Has 2 different versions – Runge’s approach and Kutta’s approach.

Rubric:

- 1 point for each valid comparison.

3. In Figure 2, consider the shaded regions A and B , portrayed beneath the two functions $f_a(x)$ and $f_b(x)$ respectively. Suppose the positive difference of the areas of the shaded regions is represented by d , which means $d = B - A$. The aforementioned functions are $f_a(x) = x^3 - 7x + 6$ and $f_b(x) = 4x$. Using the Mixed Simpson's rule i.e. Simpson's 1/3 rule (with $n_1 = 4$) and Simpson's 3/8 rule (with $n_2 = 3$), approximate the value of d .

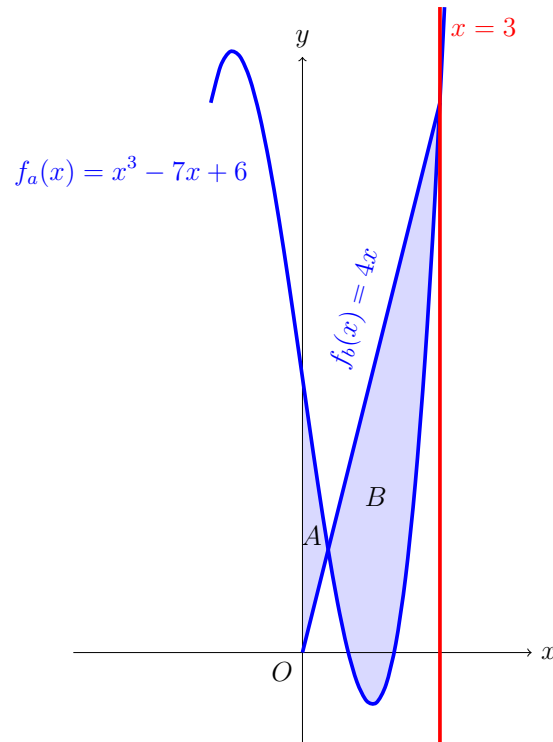


Figure 1: The shaded regions A and B .

Solution:

Suppose, the curve $f_a(x)$ and the straight line $f_b(x)$ intersect at the point where $x = p$ for the first time within the closed interval $[0, 3]$. The roots of $f_b(x) = x^3 - 7x + 6$ are -3 , 1 , and 2 , which means the curve intersects the x -axis at $x = 1$ and $x = 2$ within $[0, 3]$. Given these premises, we can posit the shaded regions A and B to be as follows,

$$A = \int_0^p (x^3 - 7x + 6) dx - \int_0^p 4x dx \quad (1)$$

$$B = \int_p^3 4x dx - \int_p^1 (x^3 - 7x + 6) dx - \int_2^3 (x^3 - 7x + 6) dx + \left(- \int_1^2 (x^3 - 7x + 6) dx \right) \quad (2)$$

Hence, we can define the difference d to be,

$$\begin{aligned}
 B - A &= \left(\int_p^3 4x \, dx - \int_p^1 (x^3 - 7x + 6) \, dx - \int_2^3 (x^3 - 7x + 6) \, dx - \int_1^2 (x^3 - 7x + 6) \, dx \right) - \\
 &\quad \left(\int_0^p (x^3 - 7x + 6) \, dx - \int_0^p 4x \, dx \right) \\
 &= \left(\int_p^3 4x \, dx + \int_0^p 4x \, dx \right) - \\
 &\quad \left(\int_p^1 (x^3 - 7x + 6) \, dx + \int_2^3 (x^3 - 7x + 6) \, dx + \int_1^2 (x^3 - 7x + 6) \, dx + \int_0^p (x^3 - 7x + 6) \, dx \right) \\
 &= \int_0^3 4x \, dx - \int_0^3 (x^3 - 7x + 6) \, dx \\
 &= \int_0^3 (-x^3 + 11x - 6) \, dx
 \end{aligned}$$

Now, we can proceed and work on approximating this integral to approximate d .

$$\text{Here, } h = \frac{b-a}{n} = \frac{b-a}{n_1+n_2} = \frac{3-0}{7} \approx 0.42857.$$

Let's compute the functional values at the right-hand side of each of these $n = 7$ segments.

$x_0 = a = 0;$	$f(x_0) = -6$
$x_1 = a + h = 0 + 0.42857;$	$f(x_1) = -1.364$
$x_2 = x_1 + h = 0.857;$	$f(x_2) = 2.7975$
$x_3 = x_2 + h = 1.285;$	$f(x_3) = 6.0131$
$x_4 = x_3 + h = 1.714;$	$f(x_4) = 7.8186$
$x_5 = x_4 + h = 2.142;$	$f(x_5) = 7.7341$
$x_6 = x_5 + h = 2.571;$	$f(x_6) = 5.2865$
$x_7 = x_6 + h = 3;$	$f(x_7) = 0$

The multiple-segment equation for Simpson's 1/3 rule is

$$I_1 = \int_a^b f(x)dx \approx \frac{b-a}{3n_1} \left[f(x_0) + 4 \sum_{\substack{i=1, \\ i=\text{odd}}}^{n_1-1} f(x_i) + 2 \sum_{\substack{i=2, \\ i=\text{even}}}^{n_1-2} f(x_i) + f(x_{n_1}) \right]$$

where, $a = 0$ and $b = x_4 = 1.714$. Plugging in the respective values, we get,

$$\begin{aligned}
 I_1 &\approx \frac{1.714}{3 \times 4} [-6 + 4 \times (-1.364 + 6.0131) + 2 \times (2.7975) + 7.8186] \\
 &= 3.7042
 \end{aligned}$$

The multiple-segment equation for Simpson's 3/8 rule is

$$I_2 = \int_a^b f(x)dx \approx \frac{3 \times (b-a)}{8n_2} \left[f(x_{n_1}) + 3 \sum_{\substack{i=n_1+1, \\ (i-n_1)\%3 \neq 0}}^{n_1+n_2-1} f(x_i) + 2 \sum_{\substack{i=n_1+3, \\ (i-n_1)\%3 = 0}}^{n_1+n_2-3} f(x_i) + f(x_{n_1+n_2}) \right]$$

where, $a = x_4 = 1.714$ and $b = x_7 = 3$. Plugging in the respective values, we get,

$$\begin{aligned}
 I_2 &\approx \frac{3 \times 1.285}{8 \times 3} [7.8186 + 3 \times (7.7341 + 5.2865) + 0] \\
 &= 7.5302
 \end{aligned}$$

So the positive difference of the shaded regions A and B is,

$$d = \int_0^3 -x^3 + 11x - 6 \, dx \approx I_1 + I_2 = 3.7042 + 7.5302 = 11.2343$$

Rubric:

- 1 point for the correct Simpson's 1/3 rule formula.
- 1.5 points for a correct I_1 value.
- 1 point for the correct Simpson's 3/8 rule formula.
- 1.5 points for a correct I_2 value.
- 1 point for an approximately correct answer.

4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a math meme related to this course. 2 (CO69) (PO420)

Warning: Cringe and lame memes will result in negative marking.

Solution:

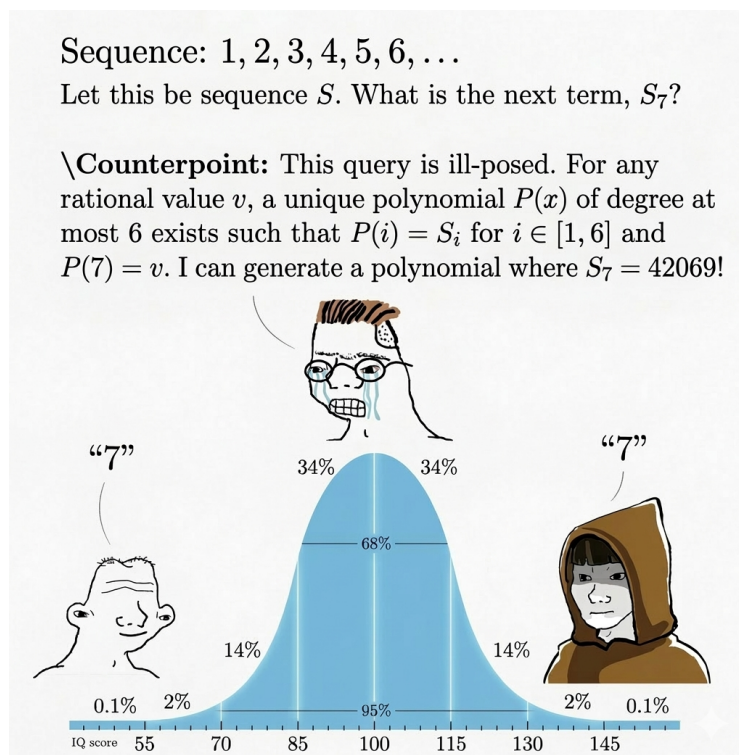


Figure 2: Occam's Razor?

Rubric:

- 2 points for a dank meme.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2024-2025
FULL MARKS: 75

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Explain how numerical approaches differ from analytical approaches to solve a problem.

4
(CO4)
(PO1)

Solution:

Any 2 valid differences will suffice. Some differences can be written as follows —

Criteria	Numerical Methods	Analytical Methods
Type of yielded solution	Approximate solution	Exact solution
Working Procedure	Keep iterating until the obtained solution is a “good enough” approximation.	Perform symbolic manipulations to arrive at an equation that yields the exact solution.
Example	Solving $ax^2 + bx + c = 0$ using the quadratic equation $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Solving $ax^2 + bx + c = 0$ using any root-finding algorithm e.g. the Bisection method.

Rubric:

- 2 points for each valid difference.

- b) Derive the formula for the Secant method geometrically.

6
(CO4)
(PO1)

Solution:

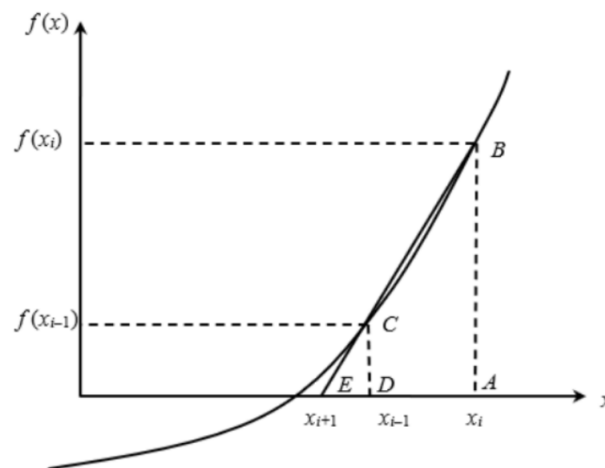


Figure 1: Geometrical representation of the Secant method.

Here, ABE and DCE are similar triangles.

Hence,

$$\begin{aligned}
 \frac{AB}{AE} &= \frac{DC}{DE} \\
 \frac{f(x_i)}{x_i - x_{i+1}} &= \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}} \\
 \Rightarrow f(x_i)(x_{i-1} - x_{i+1}) &= f(x_{i-1})(x_i - x_{i+1}) \\
 \Rightarrow f(x_i)x_{i-1} - x_{i+1}f(x_i) &= x_if(x_{i-1}) - x_{i+1}f(x_{i-1}) \\
 \Rightarrow x_{i+1}f(x_{i-1}) - x_{i+1}f(x_i) &= f(x_{i-1})x_i - f(x_i)x_{i-1} \\
 \Rightarrow x_{i+1} &= \frac{x_if(x_{i-1}) - f(x_i)x_{i-1}}{f(x_{i-1}) - f(x_i)} + x_i - x_i \\
 \Rightarrow x_{i+1} &= x_i + \frac{x_if(x_{i-1}) - f(x_i)x_{i-1} - x_if(x_{i-1}) + x_if(x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i + \frac{-f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}
 \end{aligned}$$

Hence, we get the formula for the Secant method,

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Rubric:

- 4 points for a correct geometric derivation.
- 2 points for the correct formula.

- c) Figure 2 portrays the orbital geometry of planet Earth around the Sun, where S denotes the position of the Sun and E denotes the position of planet Earth.

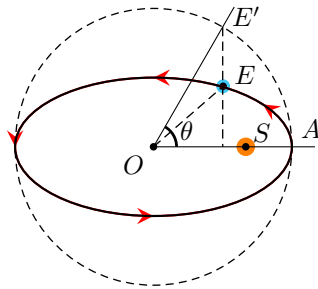


Figure 2: Geometry of the planetary orbit of Earth around the Sun for Question 1.c

Let θ denote the angle defined by $\angle E'OA$, measured in radians. The dotted circle is concentric to the ellipse and has a diameter equal to the major axis of the ellipse. Let T be the total period of planet Earth, and let t be the time it takes for Earth to go from A to E . The relationship between θ and t is dictated by Kepler's equation from orbital mechanics,

$$\theta - \epsilon \sin(\theta) = \frac{2\pi t}{T} \quad (1)$$

Here ϵ is the eccentricity of the elliptical orbit. In the case of planet Earth, the conventional value is $\epsilon = 0.017$.

- i. Formulate a function $f(\theta)$ such that its root will yield the value of the angle θ for a given value of t .

2.5
(CO1)
(PO2)

Solution:

The given equation is,

$$\theta - \epsilon \sin(\theta) = \frac{2\pi t}{T}$$

For planet Earth, the value of ϵ is 0.017, so the equation becomes,

$$\begin{aligned}\theta - 0.017 \sin(\theta) &= \frac{2\pi t}{T} \\ \Rightarrow \theta - 0.017 \sin(\theta) - \frac{2\pi t}{T} &= 0\end{aligned}$$

So the function $f(\theta)$ whose root yields the value of the angle θ for a given value of t is,

$$f(\theta) = \theta - 0.017 \sin(\theta) - \frac{2\pi t}{T}$$

Rubric:

- 2.5 points for the correct function $f(\theta)$.

- ii. Apply the Newton-Raphson method to estimate the value of θ corresponding to $t = \frac{T}{8}$. The initial guess is $\theta_0 = \pi$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 and fill it out after performing the necessary calculations.

11.5
(CO2)
(PO1)

Table 1: The relevant values obtained in the answer of Question 1.c

Iteration	θ_i	$f(\theta_i)$	$f'(\theta_i)$	θ_{i+1}	$ \epsilon_a \%$	m
1	π					
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	θ_i	$f(\theta_i)$	$f'(\theta_i)$	θ_{i+1}	$ \epsilon_a \%$	m
1	π	$\frac{3\pi}{4}$	1.017	0.824783912	280.8988764%	0
2	0.824783912	0.02690092927	0.9884618335	0.7975689722	3.412236524%	1
3	0.7975689722	$4.58439207 \times 10^{-6}$	0.9881263745	0.7975643327	$5.817059919 \times 10^{-4}$	4
4	0.7975643327	-2.0113×10^{-11}	0.988126318	0.7975643327	$2.552019839 \times 10^{-9}$	10

Note: The answers delineated in the table above are highly precise and aren't indicative of other validly approximated answers.

Rubric:

- 0.5 point for each approximately correct answer.
- 1 point for demonstrating the step-by-step procedure in each iteration.

2. a) Define Truncation Error. What are the different avenues through which Truncation Error can be introduced? Explain with proper examples.

5
(CO4)
(PO1)

Solution:

Truncation Error is the error caused by approximating the mathematical process or the modeling technique. Some of the ways it can be introduced are —

1. **Chopping off terms in a series:** If we consider only the first 3 terms of the Maclaurin series of e^x , we get $e^x \approx \left(1 + x + \frac{x^2}{2!}\right)$. And so the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= e^x - \left(1 + x + \frac{x^2}{2!}\right)$$

2. **Using a finite step size instead of an infinitesimal value:** To find the derivative of a function $f(x)$ using a finite step size h instead of an infinitesimally small h , the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= \left(\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}\right) - \left(\frac{f(x+h) - f(x)}{h}\right)$$

3. **Approximating area under curve with rectangles:** To find the integral value of a function $f(x)$ within given bounds a and b using n rectangles of width Δx_i and height $f(x_i)$, the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= \int_a^b f(x)dx - \sum_{i=0}^n f(x_i) \cdot \Delta x_i$$

Rubric:

- 1 point for a correct definition.
- 2 points for each valid example. Any 2 valid examples will suffice.

- b) The Taylor series for a function $f(x)$ is —

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of —

- i. $\ln(1+x)$
- ii. $\ln(1-x)$
- iii. $\ln\left(\frac{1+x}{1-x}\right)$
- iv. $\log_{(1-x)}(1+x)$

6+4+
4+4
(CO4)
(PO1)

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$, and $\frac{d}{dx}(x^n) = nx^{n-1}$.

- (i) Now, for $f(x) = \ln(1+x)$, let's calculate the derivatives at the point $x = 0$.

$$\begin{aligned} f(x) &= \ln(1+x), f(0) = \ln(1+0) = \ln(1) = 0; f'(x) = \frac{1}{1+x}, f'(0) = \frac{1}{1+0} = 1; \\ f''(x) &= -\frac{1}{(1+x)^2}, f''(0) = -\frac{1}{(1+0)^2} = -1; f'''(x) = \frac{2}{(1+x)^3}, f'''(0) = \frac{2}{(1+0)^3} = 2; \\ f^{(4)}(x) &= -\frac{6}{(1+x)^4}, f^{(4)}(0) = -\frac{6}{(1+0)^4} = -6; \end{aligned}$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned} f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\ \Rightarrow f(h) &= 0 + h - \frac{h^2}{2!} + \frac{2h^3}{3!} - \frac{6h^4}{4!} + \dots \\ \Rightarrow f(x) &= 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1+x) &= 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1+x) &= x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \end{aligned}$$

(ii) **Short version:** Here,

$$\begin{aligned} f(x) &= \ln(1-x) \\ &= \ln(1+(-x)) \\ &= (-x) - \frac{(-x)^2}{2!} + \frac{2(-x)^3}{3!} - \frac{6(-x)^4}{4!} + \dots \quad [\text{using the Maclaurin series of } \ln(1+x) \text{ from (i)}] \\ &= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \end{aligned}$$

Long version: Now, for $f(x) = \ln(1-x)$, let's calculate the derivatives at the point $x = 0$.

$$\begin{aligned} f(x) &= \ln(1-x), f(0) = \ln(1-0) = \ln(1) = 0; f'(x) = -\frac{1}{1-x}, f'(0) = -\frac{1}{1-0} = -1; \\ f''(x) &= -\frac{1}{(1-x)^2}, f''(0) = -\frac{1}{(1-0)^2} = -1; f'''(x) = -\frac{2}{(1-x)^3}, f'''(0) = -\frac{2}{(1-0)^3} = -2; \\ f^{(4)}(x) &= -\frac{6}{(1-x)^4}, f^{(4)}(0) = -\frac{6}{(1-0)^4} = -6; \end{aligned}$$

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$\begin{aligned} f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\ \Rightarrow f(h) &= 0 - h - \frac{h^2}{2!} - \frac{2h^3}{3!} - \frac{6h^4}{4!} + \dots \\ \Rightarrow f(x) &= 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1-x) &= 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \\ \Rightarrow \ln(1-x) &= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \end{aligned}$$

(iii) We know, $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$.

So,

$$\begin{aligned}\ln\left(\frac{1+x}{1-x}\right) &= \ln(1+x) - \ln(1-x) \\ &= \left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right) - \left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right) \\ &= 2x + \frac{4x^3}{3!} + \dots\end{aligned}$$

(iv) We know, $\log_a b = \frac{\log_c b}{\log_c a}$.

So,

$$\begin{aligned}\log_{(1-x)}(1+x) &= \frac{\ln(1+x)}{\ln(1-x)} \\ &= \frac{\left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right)}{\left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots\right)}\end{aligned}$$

Performing algebraic long division, we get,

$$\begin{array}{r} -1 + x - \frac{x^2}{2!} + \frac{5x^3}{12} + \dots \\ -x - \frac{x^2}{2!} - \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots \overline{) \quad \quad \quad} \\ \quad x + \frac{x^2}{2!} + \frac{2x^3}{3!} + \frac{6x^4}{4!} + \dots \\ \quad \quad - \frac{2x^2}{2!} - \frac{12x^4}{4!} + \dots \\ \quad \quad \quad - x^2 - \frac{x^3}{2!} - \frac{2x^4}{3!} + \dots \\ \quad \quad \quad \quad \frac{x^3}{2!} - \frac{x^4}{6} + \dots \\ \quad \quad \quad \quad \frac{x^3}{2!} + \frac{x^4}{2! \times 2!} + \dots \\ \quad \quad \quad \quad \quad \frac{5x^4}{12} + \dots \\ \quad \quad \quad \quad \quad - \frac{5x^4}{12} + \dots \\ \quad \quad \quad \quad \quad \vdots \end{array}$$

Hence, we can write,

$$\Rightarrow \log_{(1-x)}(1+x) = -1 + x - \frac{x^2}{2!} + \frac{5x^3}{12} + \dots \quad [\text{Quotient of the above algebraic long division.}]$$

Rubric:

- 6 points for the correct series of $\ln(1+x)$.
- 4 points for the correct series of $\ln(1-x)$.
- 4 points for the correct series of $\ln\left(\frac{1+x}{1-x}\right)$.
- 4 points for the correct series of $\log_{(1-x)}(1+x)$.

3. a) Compare and contrast the following interpolation methods: Direct method, Newton's Divided Difference method, and Lagrangian method.

4
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton's Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)(x - x_1) \dots (x - x_{n-1})$	$f_n(x) = \sum_{i=0}^n \left(\prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(\ A\ + \ b\))$, or $O((n+1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can be optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 2 points for each valid comparison.

- b) Suppose, there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Now, to approximate $f(x)$, you decide to use Newton's Divided Difference method of interpolation and obtain a quadratic interpolant $f_2(x)$. Derive the representation of 2nd order Newton's Divided Difference polynomial $f_2(x)$. 9
(CO4)
(PO1)

Solution:



Just click here.

Rubric:

- 1 point for writing the quadratic NDD interpolant $f_2(x)$ in the proper form.
- 1 point for correct derivation of b_0 .
- 2 point for correct derivation of b_1 .
- 4 points for correct derivation of b_2 .
- 1 point for writing the correct expanded form of the quadratic NDD interpolant $f_2(x)$.

- c) The specific heat capacity C of a substance is defined as the amount of heat that is required to raise the temperature of unit mass of that particular substance by 1 degree. Suppose, you are conducting an experiment to determine how much heat is required to bring some volume of water to its boiling point. The values of specific heat capacity C of water that you have calculated at different temperature values T are shown in Table 2. Determine the value of the specific heat at $T = 61^\circ\text{C}$ using the Direct method of interpolation and a second order polynomial. 6
(CO2)
(PO1)

Table 2: Specific heat C of water as a function of temperature T for Question 3.c

Temperature, T ($^{\circ}\text{C}$)	Specific Heat, C ($\text{Jkg}^{-1}\text{C}^{-1}$)
22	4181
42	4179
52	4186
82	4199
100	4217

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be —

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61^{\circ}\text{C}$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations —

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$

$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$

$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have —

$$\begin{aligned} \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \\ -\frac{67}{200} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix} \end{aligned}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3}T^2; \quad 42 \leq T \leq 82$$

At the point $T = 61$, we get,

$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2 \text{Jkg}^{-1}\text{C}^{-1}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 2 point for the correct coefficient values.
- 2 point for the correct answer.

- d) i. Why is $f_9(x)$ in Figure 3(b) demonstrably worse at capturing the pattern of $f(x)$ compared to $S(x)$ in Figure 3(c) ? Explain with necessary illustration(s).

4
(CO3)
(PO1)

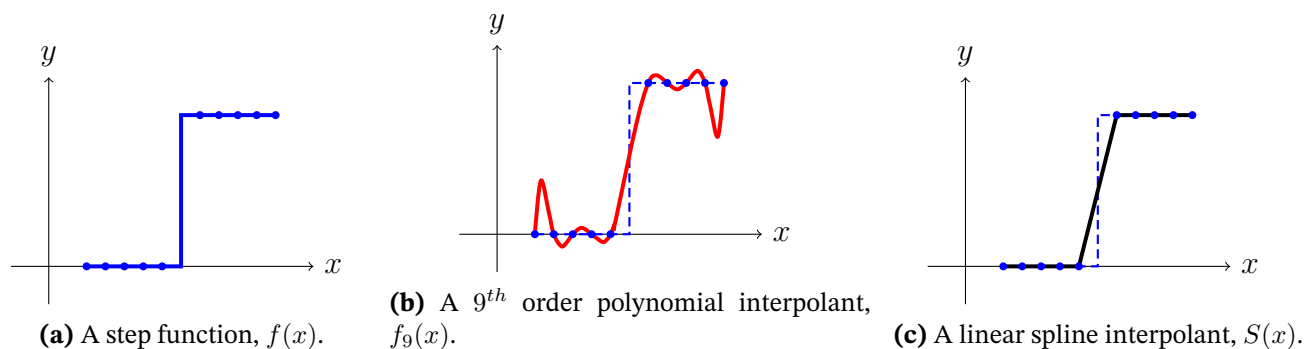


Figure 3: Predicting the nature of $f(x)$ for Question 3.d)i

Solution:

$f_9(x)$ is a worse interpolant than $S(x)$ due to the manifestation of the Runge phenomenon. Let's consider Runge's function $f(x) = \frac{1}{1 + 25x^2}$ to explain the phenomenon. When we try to approximate the nature of this function using a high-order interpolant, we observe an unhinged oscillatory behavior at either end of the interpolant.

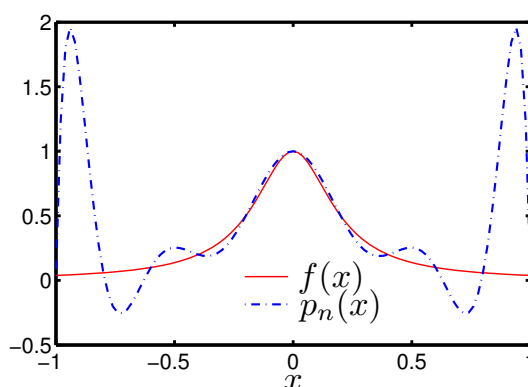


Figure 4: A 9th order polynomial interpolant passing through 10 equidistant points on Runge's function (bad).

However, we can ameliorate this phenomenon to a great extent by using splines.

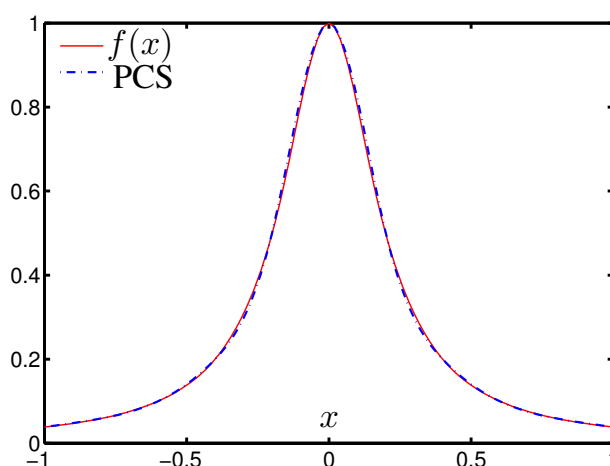


Figure 5: A 9th order cubic spline interpolant passing through 10 equidistant points on Runge's function (good).

As evident from the stark contrast between the interpolants portrayed in Figure 4 and Figure 5, splines are less susceptible to the Runge phenomenon since they stay true to the nature of the underlying function between their respective start and end points.

Rubric:

- 1 point for mentioning the correct reason.
- 3 points for illustrating nice figure(s) that explain(s) the Runge phenomenon.

- ii. Suppose you want to perform Spline Interpolation on $n + 1$ data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$ using Cubic splines of the form $f(x) = ax^3 + bx^2 + cx + d$. How many unknowns and how many equations would you have to deal with? Mention how you would obtain those equations and derive them.

5
(CO4)
(PO1)

Solution:

Each cubic spline has 4 unknowns. So, for n such splines, we have a total of $4n$ unknowns. To obtain these $4n$ unknowns, we need $4n$ equations.

Here, we have $n + 1$ given data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$. The n cubic splines that go through consecutive points and form the interpolant $f(x)$ are,

$$f(x) = \begin{cases} a_1x^3 + b_1x^2 + c_1x + d_1; & \text{for } x_0 \leq x \leq x_1 \\ a_2x^3 + b_2x^2 + c_2x + d_2; & \text{for } x_1 \leq x \leq x_2 \\ \vdots & \vdots \\ a_nx^3 + b_nx^2 + c_nx + d_n; & \text{for } x_{n-1} \leq x \leq x_n \end{cases}$$

Now, let's derive the $4n$ equations. Since the cubic splines pass through consecutive points, we get the following $2n$ equations.

$$\begin{aligned} a_1x_0^3 + b_1x_0^2 + c_1x_0 + d_1 &= f(x_0) \\ a_1x_1^3 + b_1x_1^2 + c_1x_1 + d_1 &= f(x_1) \\ \vdots & \vdots \\ a_1x_{i-1}^3 + b_1x_{i-1}^2 + c_1x_{i-1} + d_1 &= f(x_{i-1}) \\ a_1x_i^3 + b_1x_i^2 + c_1x_i + d_1 &= f(x_i) \\ \vdots & \vdots \\ a_1x_{n-1}^3 + b_1x_{n-1}^2 + c_1x_{n-1} + d_1 &= f(x_{n-1}) \\ a_1x_n^3 + b_1x_n^2 + c_1x_n + d_1 &= f(x_n) \end{aligned}$$

Now, we need to find out the 1st order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain another $n - 1$ equations in this way.

$$\begin{aligned} 3a_1x_1^2 + 2b_1x_1 + c_1 &= 3a_2x_1^2 + 2b_2x_1 + c_2 \\ \vdots & \vdots \\ 3a_{n-1}x_{n-1}^2 + 2b_{n-1}x_{n-1} + c_{n-1} &= 3a_nx_{n-1}^2 + 2b_nx_{n-1} + c_n \end{aligned}$$

Again, we need to find out the 2nd order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain yet

another $n - 1$ equations in this way.

$$\begin{array}{cccc} 6a_1x_1 + 2b_1 & = & 6a_2x_1 + 2b_2 & \\ & \vdots & \vdots & \vdots \\ 6a_{n-1}x_{n-1} + 2b_{n-1} & = & 6a_nx_{n-1} + 2b_n & \end{array}$$

Now, we require 2 more equations. For these, we consider $a_1 = 0$ and $a_{n-1} = 0$. So, in aggregate, we have $2n + (n-1) + (n-1) + 2 = 4n$ equations for the $4n$ unknowns.

Rubric:

- 1 point for the correct number of unknowns and equations.
- 1 point for the initial $2n$ equations.
- 1 point for the $n - 1$ equations using 1st order derivatives.
- 1 point for the $n - 1$ equations using 2nd order derivatives.
- 1 point for the last 2 equations.

MARKS SUMMARY TABLE

Question	Marks	Keywords
1.a	4	
1.b	6	
1.c)i	2.5	
1.c)ii	11.5	
2.a	5	
2.b	18	
3.a	4	
3.b	9	
3.c	6	
3.d)i	4	
3.d)ii	5	
Total	75.0	

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2024-2025
FULL MARKS: 150

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) What are the steps of problem-solving in engineering? With a suitable example, briefly explain how numerical methods are used to solve an engineering problem. 3 + 2
(CO2)
(PO1)
- b) What are the drawbacks of the Bisection method? Explain with proper examples and illustrations. 6
(CO4)
(PO1)
- c) Consider two ladders of known lengths $a > 0$ and $b > 0$ leaning across a straight alley of unknown width $x > 0$, touching opposite vertical walls, as portrayed in Figure 1. The triangle bounded by the ground and the lower sections of the two ladders has a known area $A > 0$. Assume the configuration is physically realizable and that $0 < x < \min(a, b)$.

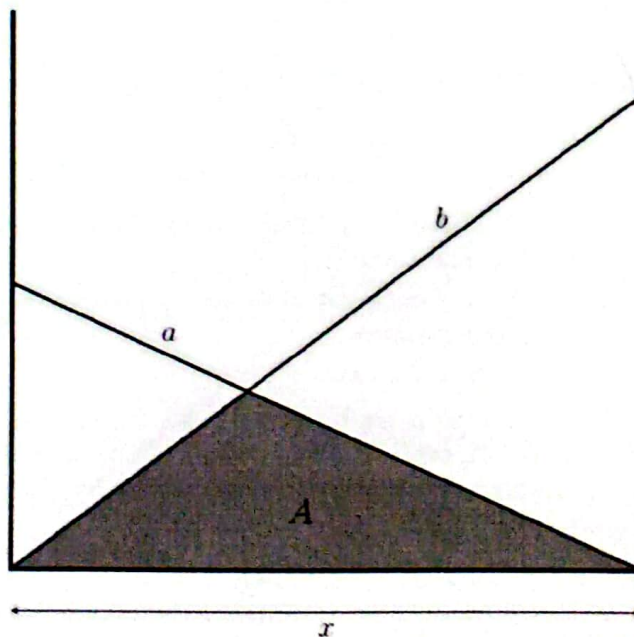


Figure 1: The crossed ladders for Question 1.c

- i. From the given scenario, formulate a function $f(x)$ (in terms of a , b , and A) whose positive root yields the width x of the alley. 12
(CO1)
(PO1)
- ii. Derive the Newton-Raphson formula for iteratively solving $f(x) = 0$, using the function f formulated in the preceding Question 1.c)i. 6
(CO2)
(PO1)

2. a) Define Round-off Error. What are the different ways through which Round-off Error can be introduced? Explain with proper examples. 5
(CO4)
(PO1)

b) The Maclaurin series for $\sin(x)$ is given as follows:

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

- i. Why is the sinusoid function $\sin(x)$ categorized as a transcendental function? 2
(CO4)
(PO1)
- ii. Use the Remainder Theorem to find bounds on the truncation error in the representation of $\sin(2)$ if only the first 3 terms of the Maclaurin series are used. 5
(CO2)
(PO1)



The Taylor series for a function $f(x)$ is —

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \operatorname{cosec}(x)$$

3. a) Prove that a polynomial of degree n or less that passes through $n+1$ data points is unique. 6
(CO4)
(PO1)

- b) i. What is the Runge phenomenon? 2
(CO3)
(PO1)

(ii) A natural cubic spline $S_3(x)$ on $[0, 2]$ is defined by:

$$S_3(x) = \begin{cases} 1 + 2x - x^3; & \text{if } 0 \leq x \leq 1, \\ 2 + a(x-1) + b(x-1)^2 + c(x-1)^3; & \text{if } 1 \leq x \leq 2. \end{cases}$$

Given that $S_3(x)$ is a natural cubic spline, it must satisfy the following conditions:

- $S_3(x)$ is continuous at the interior point $x = 1$,
- The second derivative $S_3''(x)$ vanishes at the exterior points $x = 0$ and $x = 2$, as it exhibits no curvature at those points.

Find the values of the constants a , b , and c .

- c) A robot arm with a rapid laser scanner is performing a quick quality check on holes drilled in a $15'' \times 10''$ rectangular plate. The centers of the holes describe the path the arm needs to follow. The hole centers are located on a Cartesian coordinate system (with the origin at the bottom left corner of the plate) as given in Table 1. 8
(CO2)
(PO1)

Table 1: Hole coordinates for Question 3.c

Abscissa, x (in)	2	4.25	5.25	7.81	9.2	10.6
Ordinate, y (in)	7.2	7.1	6	5	3.5	5

If the laser is traversing from $x = 2$ to $x = 4.25$ to $x = 5.25$ along a quadratic path, determine the value of y at $x = 4$ using the Newton's Divided Difference method of interpolation with a second order polynomial.

4. a) The general linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is 7+5
(CO4)
(PO1)

$$y = a_0 + a_1x$$

where a_0 and a_1 are the two parameters of the regression model.

- i. Derive the formulae for finding the optimal values of a_0 and a_1 .
- ii. Prove that the values of a_0 and a_1 obtained using the formulae derived in Question 4.a)i correspond to the absolute minimum of the optimization criterion used.
- b) In the case of the Logistic Growth model, determine the values of y that correspond to $x = 0$ and $x \rightarrow \infty$. 5
(CO3)
(PO1)
- c) Suppose, you are given n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ and you want to regress them to the Exponential model $y = e^{bx}$. 5
(CO3)
(PO2)
- Without performing any data transformation, you obtain the regression model $y = e^{k_1x}$.
 - With data transformation via linearization, you obtain the regression model $y = e^{k_2x}$.
- With proper reasoning, explain whether k_1 and k_2 are equal or not.
5. a) i. Using the method of coefficients, derive the formula for the Single-segment Trapezoidal rule. 7
(CO2)
(PO1)
- ii. Applying the formula from Question 5.a)i, derive the formula for the Multi-segment Trapezoidal rule. 6
(CO2)
(PO1)

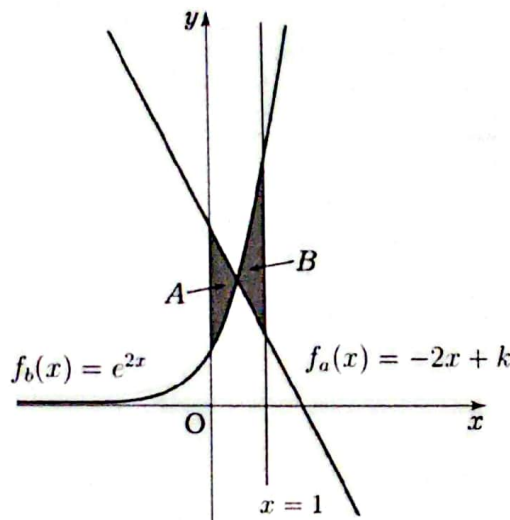


Figure 2: The shaded regions A and B for Question 5.b.

- Ⓒ In Figure 2, the area of the shaded regions A and B, portrayed beneath the two functions $f_a(x)$ and $f_b(x)$ respectively, are equal. The aforementioned functions are, 8 + 6
(CO2)
(PO1)
- $$f_a(x) = -2x + k$$
- $$f_b(x) = e^{2x}$$
- i. Using the Composite Simpson's 3/8 Rule, with $n = 9$ segments, approximate the value of k .
- ii. Analytically determine the exact value of k , denoted by k_{exact} , and use it to calculate the true error (E_t) and the absolute relative true error ($|e_t|\%$).

6. a) A ball at 1500 K is allowed to cool down in air at an ambient temperature of 500 K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by 8
(CO2)
(PO1)

$$\frac{d\theta}{dt} = -1.2 \times 10^{-12} (\theta^4 - 625 \times 10^8)$$

where θ is in K and t in seconds. Find the temperature at $t = 200$ seconds using Runge-Kutta 2nd order method (Heun's version). Assume a step size of $h = 100$ seconds.

(b)

- i. What is the LU decomposition of A, where 6
(CO2)
(PO1)

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

- ii. Determine the order and degree of each of these higher-order differential equations 0.5 × 4

(a) $x^2 \frac{dy}{dx} - \left(\frac{d^3y}{dx^3} \right)^3 + 69 = 0$

(b) $\left(\frac{dy}{dx} \right)^2 - 420xy = 3$ (CO4)
(PO1)

(c) $\frac{d^3y}{dx^3} + x^4 \frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^4 = \sin x$

(d) $\frac{dy}{dx} + xy = x^2$

- (c) Derive the orders of error (in Big O notation) for the Forward Difference rule, Backward Difference rule, and Central Difference rule of Numerical Differentiation. Show mathematical reasoning for determining the best one of the three aforementioned methods. 6 + 2
(CO2)
(PO1)